VENKATESH MUDALIAR

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SUMMARY

AI Engineer with 6+ years of engineering experience building agentic AI systems, RAG pipelines, and LLM evaluation frameworks on top of production APIs and enterprise integrations. Built Python/FastAPI and LangGraph agents with tool calling, human approval gates, and explainable decision logs. M.S. in Data Science from Stevens Institute of Technology.

TECHNICAL SKILLS

LLMs & Agentic AI: LangGraph, LangChain, RAG, Tool Calling, MCP, OpenAI APIs, Prompt Engineering, LLM Evaluation, Human-in-the-Loop Workflows, FAISS, Embeddings
ML & Data: scikit-learn, Hugging Face, pandas, NumPy, PySpark, Feature Engineering, Model Evaluation, PostgreSQL
Languages & Backend: Python, TypeScript, JavaScript, Java, SQL, FastAPI, REST APIs, Node.js
Frontend: React, Next.js, Chrome Extension APIs
Cloud & Dev Tools: AWS, GCP, Git, GitHub, Docker, pytest, CI/CD

PROFESSIONAL EXPERIENCE

AI Engineer | MarketEq Digital, Remote **Jun 2026 - Present**

- Built an autonomous lead-sourcing **AI agent** (**Python, FastAPI**) using a **Thompson-sampling** bandit to decide which of 100+ live data sources to test, pause, or scale within budget limits, with safeguards against noisy or volume-only signals.
- Implemented **LLM-driven approvals** for the orchestrator agent: an LLM judges each suggestion against business goals after a budget check, with every decision logged in plain language and controlled by a user setting.
- Validated agents on live **PostgreSQL** data (29.5K companies across two GCP Cloud SQL databases) using read-only diagnostics, covered by **319 automated pytest tests** and shipped through GitHub pull requests.

Full Stack Software Engineer | Easley Dunn Productions, SF Bay Area, CA **Jun 2026 - Present**

- Developed and debugged application features spanning **Unity/C#** gameplay logic, application data, and event workflows, validating changes with automated and manual test cases.
- Applied **AI-assisted development** workflows to investigate complex code paths and deliver reviewed **GitHub pull requests**.

AI/ML Research Assistant | Stevens Institute of Technology, Hoboken, NJ **Apr 2025 - May 2026**

- Designed **ML evaluation pipelines** benchmarking traditional models against **LLM-based forecasting** on financial datasets, improving **F1-score 12%** via cross-validated feature selection.
- Built **FAISS**-based semantic search and retrieval-augmented pipelines integrating multi-source financial data, improving **retrieval precision 18%** across LLM benchmarking experiments.

Software Engineer - Enterprise Integration | Accenture, India **Feb 2023 - Sep 2024**

- Spearheaded a large-scale ERP migration, modernizing **50+ SAP integration interfaces** from legacy PI/PO middleware to SAP's cloud-native **CPI** platform, cutting data-processing time **30%**.
- Built cloud-based **API and integration pipelines** connecting SAP S/4HANA with Salesforce, SuccessFactors, Ariba, and external systems using REST, OData, SOAP, and IDoc.

Software Engineer - Enterprise Integration | LTIMindtree Ltd., India **Sep 2020 - Feb 2023**

- Built a **full-stack integration monitoring dashboard** using React, TypeScript, FastAPI, and PostgreSQL to centralize interface health, failure trends, and error details, reducing troubleshooting time by **45%**.
- Led end-to-end **automation** of Sourcing and Procurement workflows in **SAP S/4HANA Cloud**, replacing manual processes and cutting manual intervention **60%**.

AI/ML PROJECTS

SAP IntelliOps: Multi-Agent Incident Resolution | GitHub

- Built an agentic incident-resolution system that investigates SAP CPI failures using **LangGraph**-orchestrated agents, **RAG**, and **tool calling**, with policy-gated execution and **human approval** for higher-risk remediation.
- Evaluated the system across incident classification, remediation, and escalation scenarios, achieving **20/20** taxonomy accuracy, **16/16** known-remediation accuracy, and **1.00** escalation F1.

JobApply Copilot: AI-Powered Job Application Platform | GitHub

- Designed and deployed JobApply Copilot end to end using Next.js, React, TypeScript, FastAPI, PostgreSQL, and OpenAI APIs, with a **Chrome extensio**n for job extraction and AI-generated application materials.
- Built a searchable job-tracking **dashboard** with status filters, activity charts, **REST API** workflows, and AI-generated outreach messages; deployed the application on Render.

EDUCATION

M.S. Data Science ***Sep 2024 - May 2026***
Stevens Institute of Technology | GPA 3.82 *Hoboken, NJ*

B.E. Electronics & Telecommunications ***Apr 2016 - May 2020***
University of Mumbai *Mumbai, India*

CERTIFICATIONS & AWARDS

AWS Certified AI Practitioner | IBM Data Science Professional Certificate | Databricks Generative AI Fundamentals
Awards: GoMax Award, LTIMindtree | Best Performer of the Month, Accenture